1. A radio wave is traveling the negative y-direction. What is the direction of $\vec{E}$ at a point where $\vec{B}$ is in the positive x-direction?

2. A radio antenna broadcasts a 1.0 MHz radio wave with 25 kW of power. Assume that the radiation is emitted uniformly in all directions.

   A. What is the wave's intensity 30 km from the antenna?

   B. What is the electric field amplitude at this distance?

3. A lightwave has a peak electric field of 1000 N/C, initially pointing in the +y direction. The magnetic field is initially in the +z direction.

   A. What is the direction of propagation of the light wave?

   B. What is the peak amplitude of the magnetic field?

   C. What the intensity of the light wave?

   D. How much radiation pressure does the wave carry?

   E. How much force would the plane wave exert on a $8\frac{1}{2}^{"} \times 11^{"}$ piece of paper normal to it?

4. The Sun's Luminosity, or power, is $L_\odot = 3.84 \times 10^{26}$ Watts. Use **GeoGebra** to:

   A. Determine the Intensity of the sun at the Earth, 1 AU away.

   B. Determine the Intensity of the sun at the Jupiter 5.2 AU away.

   C. Plot the sun's intensity as a function of distance, in AU. Add a slider for the distance and a point that rides along the curve.

   D. Use **Intersect** to find how many AU's from the sun, the intensity will be $I = 1.00 \frac{\text{Watts}}{\text{m}^2}$

   E. What is the potential maximum power budget of Earth's biosphere, assuming that plants are able to convert 100% of the incident intensity into useful power?

   F. How does this compare to humanities yearly energy use of 5×10^{20} Joules?

GeoGebra hints

Type each line into the Input Bar. Anything you define stays in the Algebra view and can be reused below. Do **not** name the function `I` — GeoGebra reserves `i`.

```
L = 3.84 * 10^26                g(x) = 1
AU = 1.496 * 10^11              Intersect(Isun, g)
Isun(r) = L / (4 pi (r AU)^2)
r = Slider(0.1, 40, 0.1)        Rearth = 6.371 * 10^6
P = (r, Isun(r))                Pearth = Isun(1) * pi * Rearth^2
```

`Isun(1)` is the intensity at Earth. To see the curve type `Isun(x)`, then right-click the Graphics view → **Graphics...** and set x from 0 to 40, y from 0 to 1500. `Intersect` finds where that curve crosses the line $I = 1$ — you can also just pick up the **Intersect** tool and click the two curves. For **E**, remember Earth intercepts sunlight over its *cross-section*: a disk of radius $R_\oplus = 6.371 \times 10^6$ m, not its whole surface.

5. Find the frequency of the Hα, and Hβ spectal lines, which have a wavelengths of 656.3, and 486.0 nm respectively.

6. Build a GeoGebra tool that finds the frequency that corresponds to any wavelength. You will need to create a function to accomplish this. Add a slider for the wavelength and a dynamic text that reports the matching frequency, so you can read off any wavelength you like.

7. What are the wavelength and freqency ranges of radio, microwave, infrared, optical, UV, x-ray, and gamma radiation?

